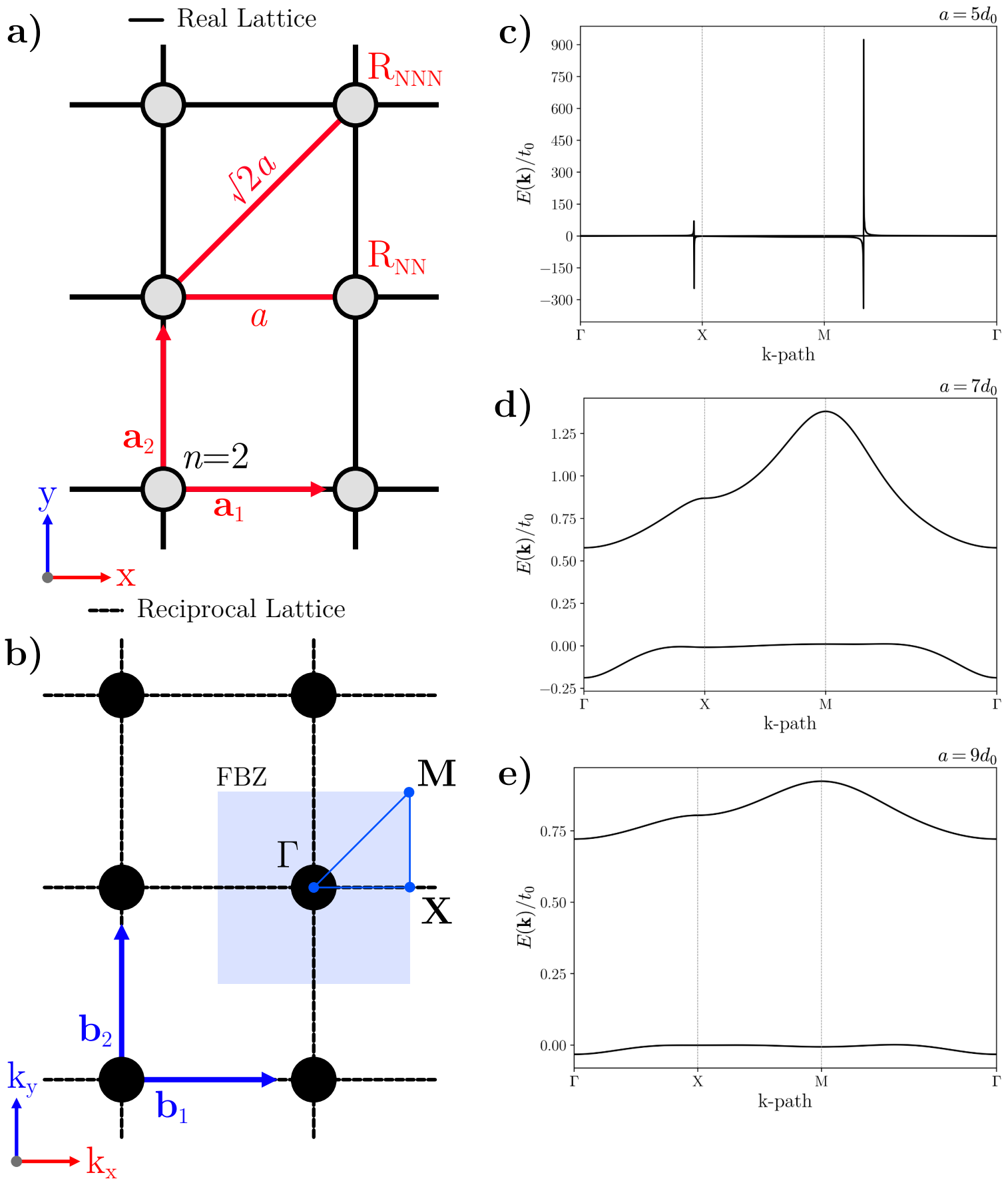

a)
Real Lattice
R_{NNN}
$\sqrt{2}a$
R_{NN}
a
$\mathbf{a}_2$
n=2
$\mathbf{a}_1$
y
x
Reciprocal Lattice
b)
FBZ
M
Γ
X
$\mathbf{b}_2$
$\mathbf{b}_1$
k_y
k_x
c)
$a = 5d_0$
$E(\mathbf{k})/t_0$
900
750
600
450
300
150
0
−150
−300
Γ
X
M
Γ
k-path
d)
$a = 7d_0$
$E(\mathbf{k})/t_0$
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
Γ
X
M
Γ
k-path
e)
$a = 9d_0$
$E(\mathbf{k})/t_0$
0.75
0.50
0.25
0.00
Γ
X
M
Γ
k-path